CLASS 8 · JEE MAINS · INSTITUTE TOP 10 VS OUR TOP 10

Pragya G8 (EXI) Achievers

Part Test-1 · where our toppers fall behind the institute's toppers

Exam date 08 Oct 2026
Campus HYD/Nallakunta DD Colony
Compared institute top 10 vs our top 10

On average the institute's top 10 scored **224.3** and our top 10 scored **169.8** – a gap of **54.5 marks** per student: **3.0** in Maths, **30.7** in Physics and **20.8** in Chemistry.

Physics 56% of the gap | Chemistry 38% of the gap

Scores are rebuilt from each question's right / wrong / skipped counts with the paper's marking (our average matches the official 169.8 exactly).

224 / 296
institute top 10 – average score

170 / 296
our top 10 – average score

12 questions
where 4+ more of the institute top 10 got it right

4 questions
where 2+ more of ours got it right

1 Subject by subject

Average marks per student, and how the answers split into right, wrong and skipped. ■ Institute top 10 ■ Our top 10

SUBJECT	AVERAGE SCORE		GAP	RIGHT	WRONG	SKIPPED
Maths out of 96 = 100 minus Q20 (4)		65.0 62.0	3.0 per student	71% vs 69%	12% vs 18%	17% vs 12%
Physics out of 100		69.0 38.3	30.7 per student	73% vs 46%	17% vs 31%	10% vs 23%
Chemistry out of 100		90.3 69.5	20.8 per student	92% vs 74%	5% vs 20%	3% vs 6%

"Out of" = the subject's marks on this paper minus the questions left out by the final key (marks of each left-out question in brackets). Percentages are of all answers in that subject (questions × students), institute first, ours second.

Key takeaways

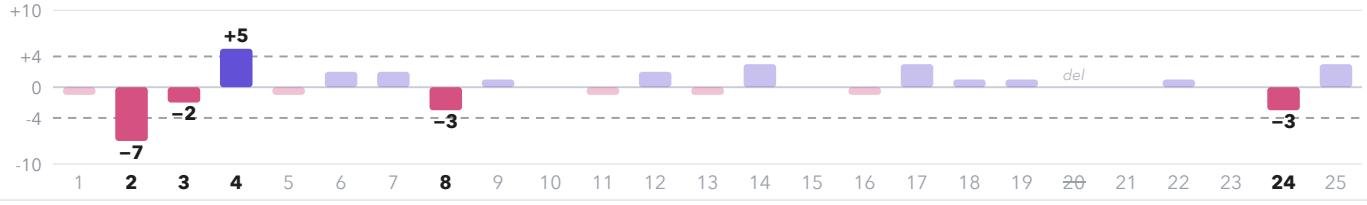
1. **Physics** is where we fall furthest behind: **30.7** marks per student (56% of the total gap).
2. **12 questions** were answered correctly by at least 4 more of the institute's top 10 than ours. Biggest: Q35 (9 vs 1), Q43 (7 vs 1), Q4 (8 vs 3).
3. Our top 10 skipped far more in **Physics**: 23% of answers left blank, against 10% for the institute's top 10.
4. We also got more wrong in **Chemistry**: 20% of answers wrong vs 5% – accuracy, not just coverage.
5. Where we did better: Q2 (10 vs 3), Q8 (10 vs 7), Q24 (8 vs 5), Q3 (5 vs 3).

2 Question-by-question gap

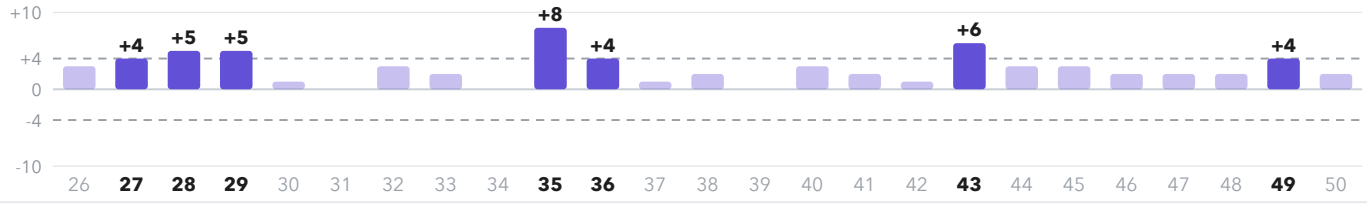
For each question: how many more of the institute's top 10 got it right than ours. Bars above the line = the institute did better; below = we did better. Gaps of 4+ (dashed line) are labelled.

Institute top 10 got more right · Our top 10 got more right · ---- Gap of 4 · "del" / "n/a" = left out (final key)

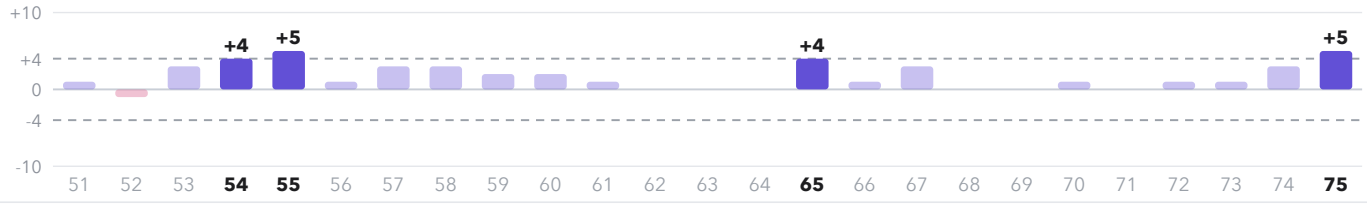
Maths — **1** question with a gap of 4+ · gap **3.0** marks per student



Physics — **7** questions with a gap of 4+ · gap **30.7** marks per student



Chemistry — **4** questions with a gap of 4+ · gap **20.8** marks per student



3 Chapter gaps

Average marks per student on each chapter's questions, institute top 10 vs ours. Subjects are ordered from the biggest gap to the smallest; within each subject, largest chapter gap first.

CHAPTER	QS	INSTITUTE VS OURS	GAP PER STUDENT	GAP QUESTIONS
1. Physics gap **30.7** marks per student · **7** questions with a gap of 4+				
Kinematics	10	28.1 vs 16.9	11.2	Q35, Q36, Q49
Laws of Motion	8	17.8 vs 7.7	10.1	Q43
Vectors	7	23.1 vs 13.7	9.4	Q27, Q28, Q29
2. Chemistry gap **20.8** marks per student · **4** questions with a gap of 4+				
Mole Concept & Basics	9	34.4 vs 27.3	7.1	Q54, Q55
Chemical Bonding	6	17.3 vs 11.9	5.4	Q75
Atomic Structure	5	19.5 vs 15.1	4.4	–
Periodic Classification	5	19.1 vs 15.2	3.9	Q65
3. Maths gap **3.0** marks per student · **1** question with a gap of 4+				
Sequences & Series	5	13.3 vs 11.8	1.5	Q4
Surds & Indices	11	35.1 vs 33.9	1.2	–
Logarithms & Exponents	3	8.5 vs 7.7	0.8	–
Quadratic Equations	5	8.1 vs 8.6	−0.5	–

4 Biggest gaps – question list

Questions where at least 4 more of the institute's top 10 got the right answer than ours, one table per subject. The cropped question for each is in section 5.

Maths

1 question with a gap of 4+

Q	CHAPTER – WHAT IT TESTS	TYPE	INSTITUTE RIGHT	OURS RIGHT · WRONG · SKIPPED	GAP
Q4	**Sequences & Series** – Identify AP / GP / HP from a surd expression	Single correct · +4 / −1	8/10	3/10 · 5 · 2	+5

Physics

7 questions with a gap of 4+

Q	CHAPTER – WHAT IT TESTS	TYPE	INSTITUTE RIGHT	OURS RIGHT · WRONG · SKIPPED	GAP
Q35	**Kinematics** – a = kv² – integrate to get v(s)	Single correct · +4 / −1	9/10	1/10 · 1 · 8	+8
Q43	**Laws of Motion** – Normal reaction across an inclined contact	Single correct · +4 / −1	7/10	1/10 · 7 · 2	+6
Q28	**Vectors** – Resultant ⊥ smaller vector; find magnitudes	Single correct · +4 / −1	10/10	5/10 · 4 · 1	+5
Q29	**Vectors** – Angles between three vectors summing to zero	Single correct · +4 / −1	10/10	5/10 · 4 · 1	+5
Q27	**Vectors** – Vector components under axis rotation	Single correct · +4 / −1	7/10	3/10 · 2 · 5	+4
Q36	**Kinematics** – a(t) with modulus – area under a-t	Single correct · +4 / −1	5/10	1/10 · 9 · 0	+4
Q49	**Kinematics** – Free fall then deceleration – find height	Numerical · +4 / −1	5/10	1/10 · 2 · 7	+4

Chemistry

4 questions with a gap of 4+

Q	CHAPTER – WHAT IT TESTS	TYPE	INSTITUTE RIGHT	OURS RIGHT · WRONG · SKIPPED	GAP
Q55	**Mole Concept & Basics** – Laws of chemical combination	Single correct · +4 / −1	10/10	5/10 · 4 · 1	+5
Q75	**Chemical Bonding** – Hybrid orbitals + lone pairs in NH_3	Numerical · +4 / −1	8/10	3/10 · 4 · 3	+5
Q54	**Mole Concept & Basics** – Minimum molecular mass of insulin from 3.4% S	Single correct · +4 / −1	10/10	6/10 · 1 · 3	+4
Q65	**Periodic Classification** – Largest radius among isoelectronic ions	Single correct · +4 / −1	8/10	4/10 · 5 · 1	+4

Where our top 10 did better

Q2 10 vs 3 right · **Q8** 10 vs 7 right · **Q24** 8 vs 5 right · **Q3** 5 vs 3 right

How to read this report. "Institute top 10" = the *Top 10 Students* rows of the institute's question-wise R/W/U sheet for this test (program ranks 1–10 in Pragya G8 (EXI) Achievers, all campuses). "Our top 10" = this campus's top 10 by program rank. A gap of +5 on a question means 5 more of the institute's top 10 got it right than ours.

Final key applied. Deleted in the final key: Q20. Answer revised in the final key and counted as marked in the final marks sheet: Q49 (key 298), Q67 (key D). Left-out questions are marked "del" or "n/a" in the charts and excluded from both groups.

5 Maths – biggest gaps

1 question, largest gap first
cropped from the question paper

Q4

Sequences & Series – **Identify AP / GP / HP from a surd expression**

Institute top 10: **8 right**, 2 wrong, 0 skipped · Our top 10: **3 right**, 5 wrong, 2 skipped

Maths · Single correct · marking +4 / −1

Gap +5

4. If a, b, c are in A.P., then $\frac{1}{\sqrt{a}+\sqrt{b}}, \frac{1}{\sqrt{a}+\sqrt{c}}, \frac{1}{\sqrt{b}+\sqrt{c}}$ are in

(A)A.P. (B)G.P. (C)H.P. (D)Neither A.P nor G.P nor H.P

Ours wrong: Kolukula Hema Shiva Naga Hariharan, M Charan Kumar Reddy, Kavuru Sahithi, Fareeha Shaik, Madipadiga Dheeraj

Ours skipped: M Jagadeesear Reddy, Aleti Dhanu Sai Sagar

Physics – biggest gaps

7 questions, largest gap first
cropped from the question paper

Q35

Kinematics – a = kv² – integrate to get v(s)

Institute top 10: **9 right**, 0 wrong, 1 skipped · Our top 10: **1 right**, 1 wrong, 8 skipped

Physics · Single correct · marking +4 / −1

Gap +8

35. If a particle accelerates with $a = kv^2$ and initial velocity = u, then find velocity after displacement s

(A) $v = ue^{ks}$ (B) $v = u^{ks}e^{s}$ (C) $v = u^{k}e^{s}$ (D) $v = eu^{ks}$

Ours wrong: Nakka Mahidhar

Ours skipped: Kolukula Hema Shiva Naga Hariharan, Ruperao Shritan, Kavuru Sahithi, Fareeha Shaik, M Jagadeesear Reddy, Karthikeya Silver, Aleti Dhanu Sai Sagar, Madipadiga Dheeraj

Q43

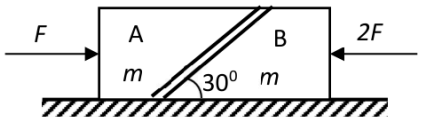
Laws of Motion – Normal reaction across an inclined contact

Institute top 10: **7 right**, 3 wrong, 0 skipped · Our top 10: **1 right**, 7 wrong, 2 skipped

Physics · Single correct · marking +4 / −1

Gap +6

43. Two blocks 'A" and 'B' each of mass 'm' are placed on a smooth horizontal surface. Two horizontal forces F and 2F are applied on both the blocks 'A' and 'B' respectively as shown in figure. The block A does not slide on block B. Then the normal reaction acting between the two blocks is



(A) F (B) $F/2$ (C) $F/\sqrt{3}$ (D) 3F

Ours wrong: Kolukula Hema Shiva Naga Hariharan, Kavuru Sahithi, Fareeha Shaik, M Jagadeesear Reddy, Karthikeya Silver, Aleti Dhanu Sai Sagar, Madipadiga Dheeraj

Ours skipped: Ruperao Shritan, Nakka Mahidhar

Q28

Vectors – Resultant ⊥ smaller vector; find magnitudes

Institute top 10: **10 right**, 0 wrong, 0 skipped · Our top 10: **5 right**, 4 wrong, 1 skipped

Physics · Single correct · marking +4 / −1

Gap +5

28. The simple sum of two co-initial vectors is 16 units . Their vector sum is 8 units. The resultant of the vectors is perpendicular to the smaller vector . The magnitudes of the two vectors are

(A) 22units and 14 units (B) 4 units and 12 units
(C) 6 units and 10 units (D) 8 units and 12 units

Ours wrong: Kavuru Sahithi, Fareeha Shaik, Nakka Mahidhar, Madipadiga Dheeraj

Ours skipped: M Jagadeesear Reddy

Q29 **Vectors – Angles between three vectors summing to zero**
Institute top 10: **10 right**, 0 wrong, 0 skipped · Our top 10: **5 right**, 4 wrong, 1 skipped
Physics · Single correct · marking +4 / −1
Gap +5

29. Given: $\vec{a}+\vec{b}+\vec{c}=0$. Out of the three vectors $\vec{a},\vec{b}$ and $\vec{c}$, two are equal in magnitude. The magnitude of the third vector is $\sqrt{2}$ times that of either of the two having equal magnitude. The angles between the vector are

(A) $90^0,135^0,135^0$ (B) $30^0,60^0,90^0$

(C) $45^0,45^0,90^0$ (D) $45^0,60^0,90^0$

Ours wrong: Kolukula Hema Shiva Naga Hariharan, M Jagadeesear Reddy, Nakka Mahidhar, Madipadiga Dheeraj

Ours skipped: Aleti Dhanu Sai Sagar

Q27 **Vectors – Vector components under axis rotation**
Institute top 10: **7 right**, 0 wrong, 3 skipped · Our top 10: **3 right**, 2 wrong, 5 skipped
Physics · Single correct · marking +4 / −1
Gap +4

27. The components of a vector along x and y directions are (n+1) and 1 respectively . If the co-ordinate system is rotated by an angle $\theta=60^0$, then the components change to n and 3. The value of n is

(A) 2 (B) $\cos 60^0$ (C) $\sin 60^0$ (D) 3.5

Ours wrong: Kavuru Sahithi, Nakka Mahidhar

Ours skipped: Ruperao Shritan, Fareeha Shaik, M Jagadeesear Reddy, Karthikeya Silver, Aleti Dhanu Sai Sagar

Q36 **Kinematics – a(t) with modulus – area under a-t**
Institute top 10: **5 right**, 4 wrong, 1 skipped · Our top 10: **1 right**, 9 wrong, 0 skipped
Physics · Single correct · marking +4 / −1
Gap +4

36. Starting from rest a particle moves in a straight line with acceleration $a=\{2+|t-2|\}\,m/s^2$. Velocity of the particle at the end of 4s will be

(A) 16 m/s (B) 20 m/s (C) 8 m/s (D) 12 m/s

Ours wrong: Kolukula Hema Shiva Naga Hariharan, Ruperao Shritan, Kavuru Sahithi, Fareeha Shaik, M Jagadeesear Reddy, Nakka Mahidhar, Karthikeya Silver, Aleti Dhanu Sai Sagar, Madipadiga Dheeraj

Ours skipped: –

Q49 **Kinematics – Free fall then deceleration – find height**
Institute top 10: **5 right**, 2 wrong, 3 skipped · Our top 10: **1 right**, 2 wrong, 7 skipped
Physics · Numerical · marking +4 / −1
Gap +4

49. A parachutist after bailing out falls 50m without friction. When parachute opens, it decelerates at $2m/\sec^2$. He reaches the ground with a speed of 3m/s. The height he bail out is nearly (in m) (Take g=10m/sec^2) .

Ours wrong: M Charan Kumar Reddy, Fareeha Shaik

Ours skipped: Ruperao Shritan, Kavuru Sahithi, M Jagadeesear Reddy, Nakka Mahidhar, Karthikeya Silver, Aleti Dhanu Sai Sagar, Madipadiga Dheeraj

Chemistry – biggest gaps

4 questions, largest gap first
cropped from the question paper

Q55

Mole Concept & Basics – **Laws of chemical combination**

Institute top 10: **10 right**, 0 wrong, 0 skipped · Our top 10: **5 right**, 4 wrong, 1 skipped

Chemistry · Single correct · marking +4 / −1

Gap +5

55. 12 g carbon combines with 64 g sulphur to form CS_2. 12 g carbon also combines with 32 g oxygen to form CO_2. 10 g sulphur combines with 10 g oxygen to form SO_2. These data illustrate the -

(A) Law of multiple proportions

(B) Law of definite proportions

(C) Law of reciprocal proportions

(D) Law of gaseous volumes.

Ours wrong: Kavuru Sahithi, Fareeha Shaik, M Jagadeesear Reddy, Karthikeya Silver

Ours skipped: Ruperao Shritan

Q75

Chemical Bonding – **Hybrid orbitals + lone pairs in NH_3**

Institute top 10: **8 right**, 1 wrong, 1 skipped · Our top 10: **3 right**, 4 wrong, 3 skipped

Chemistry · Numerical · marking +4 / −1

Gap +5

75. According to VBT, consider the molecule NH_3. The central nitrogen atom undergoes hybridisation before bond formation.

If the number of hybrid orbitals formed is x, and the number of lone pairs present on nitrogen is y, find:

x + y

Ours wrong: Kolukula Hema Shiva Naga Hariharan, Ruperao Shritan, Nakka Mahidhar, Karthikeya Silver

Ours skipped: M Charan Kumar Reddy, M Jagadeesear Reddy, Madipadiga Dheeraj

Q54

Mole Concept & Basics – **Minimum molecular mass of insulin from 3.4% S**

Institute top 10: **10 right**, 0 wrong, 0 skipped · Our top 10: **6 right**, 1 wrong, 3 skipped

Chemistry · Single correct · marking +4 / −1

Gap +4

54. Insulin contains 3.4% sulphur. What will be the minimum molecular weight of insulin ?

(A) 94.176 (B) 1884 (C) 941.176 (D) 976

Ours wrong: Karthikeya Silver

Ours skipped: Fareeha Shaik, M Jagadeesear Reddy, Nakka Mahidhar

Q65

Periodic Classification – **Largest radius among isoelectronic ions**

Institute top 10: **8 right**, 1 wrong, 1 skipped · Our top 10: **4 right**, 5 wrong, 1 skipped

Chemistry · Single correct · marking +4 / −1

Gap +4

65. Which of the following has largest radius ?

(A) Mg^{2+} (B) Na^{+} (C) O^{2-} (D) F^{-}

Ours wrong: Kolukula Hema Shiva Naga Hariharan, M Charan Kumar Reddy, Fareeha Shaik, Nakka Mahidhar, Karthikeya Silver

Ours skipped: M Jagadeesear Reddy